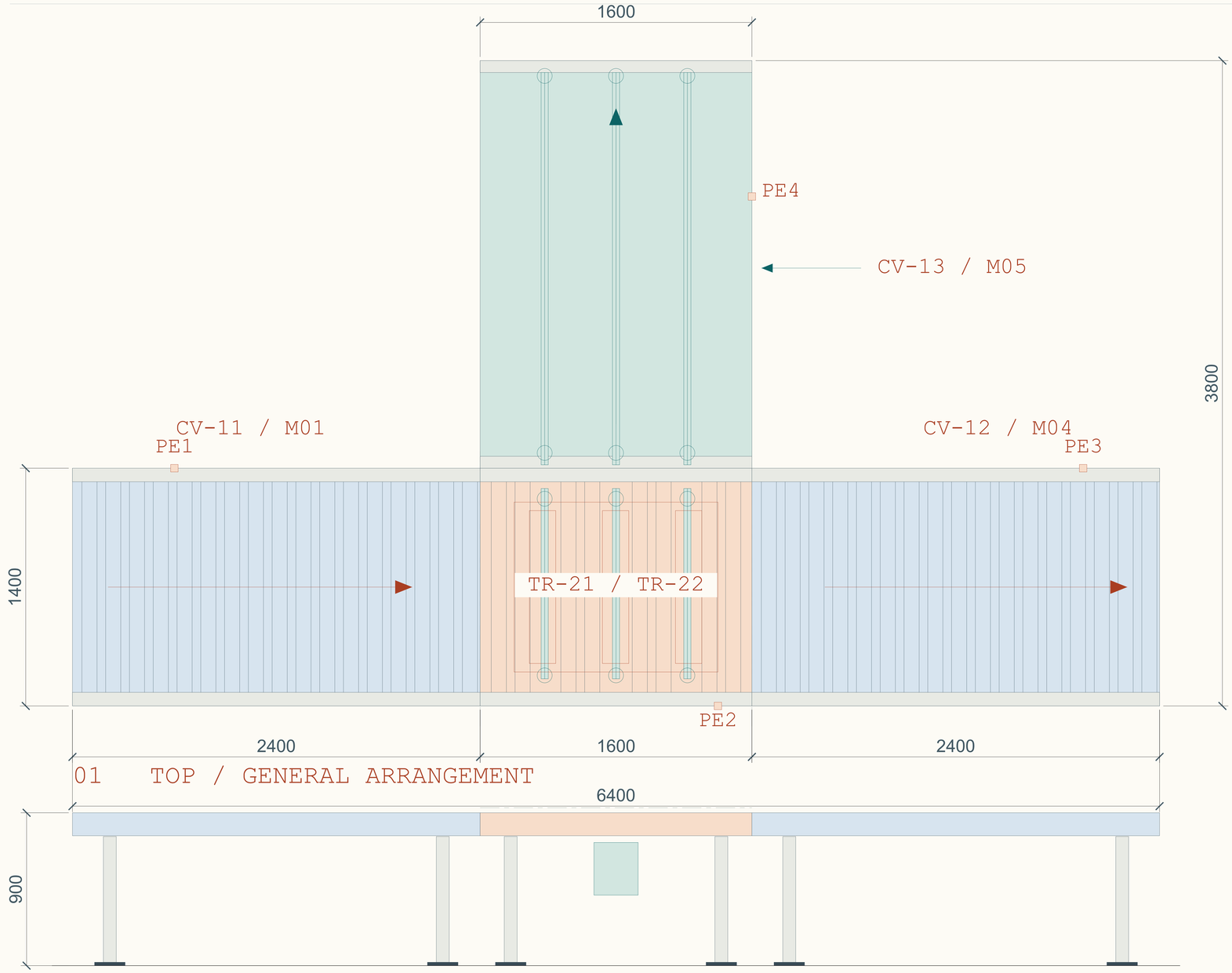
System master drawing

SYSTEM GA / INSTALLATION DATUMS / MECHANICAL, PNEUMATIC AND CONTROL DRAWING REGISTER



01 TOP / GENERAL ARRANGEMENT

02 MAIN RUN / INSTALLATION ELEVATION

REV 01 / DRAWING REGISTER

CODE	DRAWING
G000	System master drawing
M000	Equipment overview
M100	General arrangement
M110	Installation elevations
M200	Guided lift assembly
M210	Transfer drive shaft
M220	Welded base frame
P300	Lift pneumatics
C400	Operating sequence
M120	Anchor coordinate plan
M230	Chain and take-up assembly
M300	Equipment bill of materials
M400	Guarding and service layout
E500	Controls and field interfaces
Q600	Verification and handover

SYSTEM DESIGN BASIS

Pallet: 1,200 × 1,000 mm / 500 kg assumed
Roller top +900 / raised chain and side deck +930
Working lift 45 mm / three chain strands

INSTALLATION REFERENCES

Anchor coordinates and typical hole pattern: M120 / M220
Safeguard and service envelope: M400
Power, air and control interfaces: E500 / P300

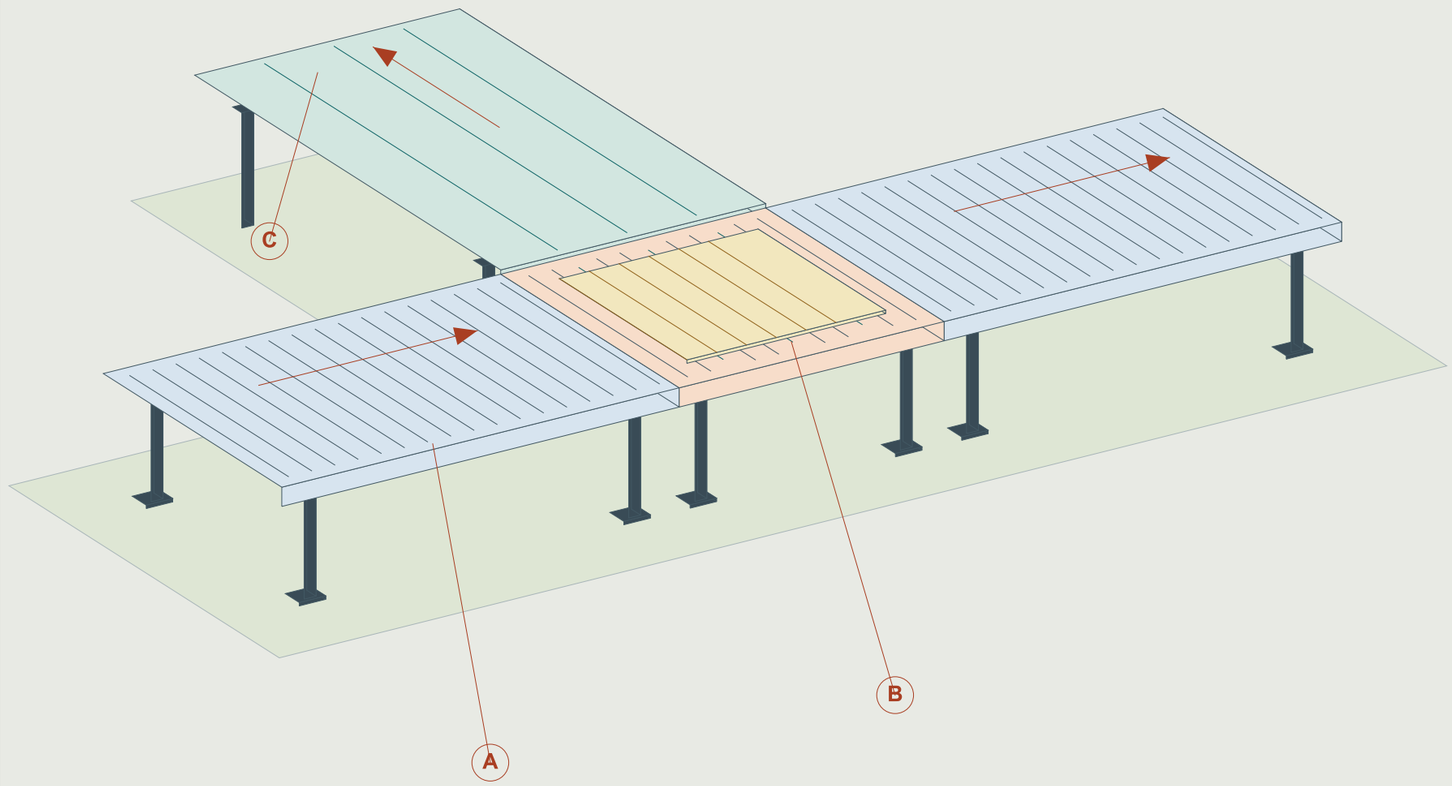
ISSUE STAGE

Coordinated concept / release checks on Q600

Equipment overview

PALLET HANDLING EQUIPMENT / PASS THROUGH OR TRANSFER THROUGH 90°

01 EQUIPMENT AXONOMETRIC



A / INFEED ROLLERS B / GUIDED TRANSFER C / SIDE DISCHARGE

SHARED EQUIPMENT DATUMS / MODULE IDS FOLLOW M100

CONFIGURATION

6,400 × 3,800

MM / OVERALL PLAN ENVELOPE

SHEET	DRAWING INDEX
M000	Equipment overview
M100	General arrangement
M110	Installation elevations
M200	Guided lift assembly
M210	Transfer drive shaft
M220	Welded base frame
P300	Lift pneumatics
C400	Operating sequence

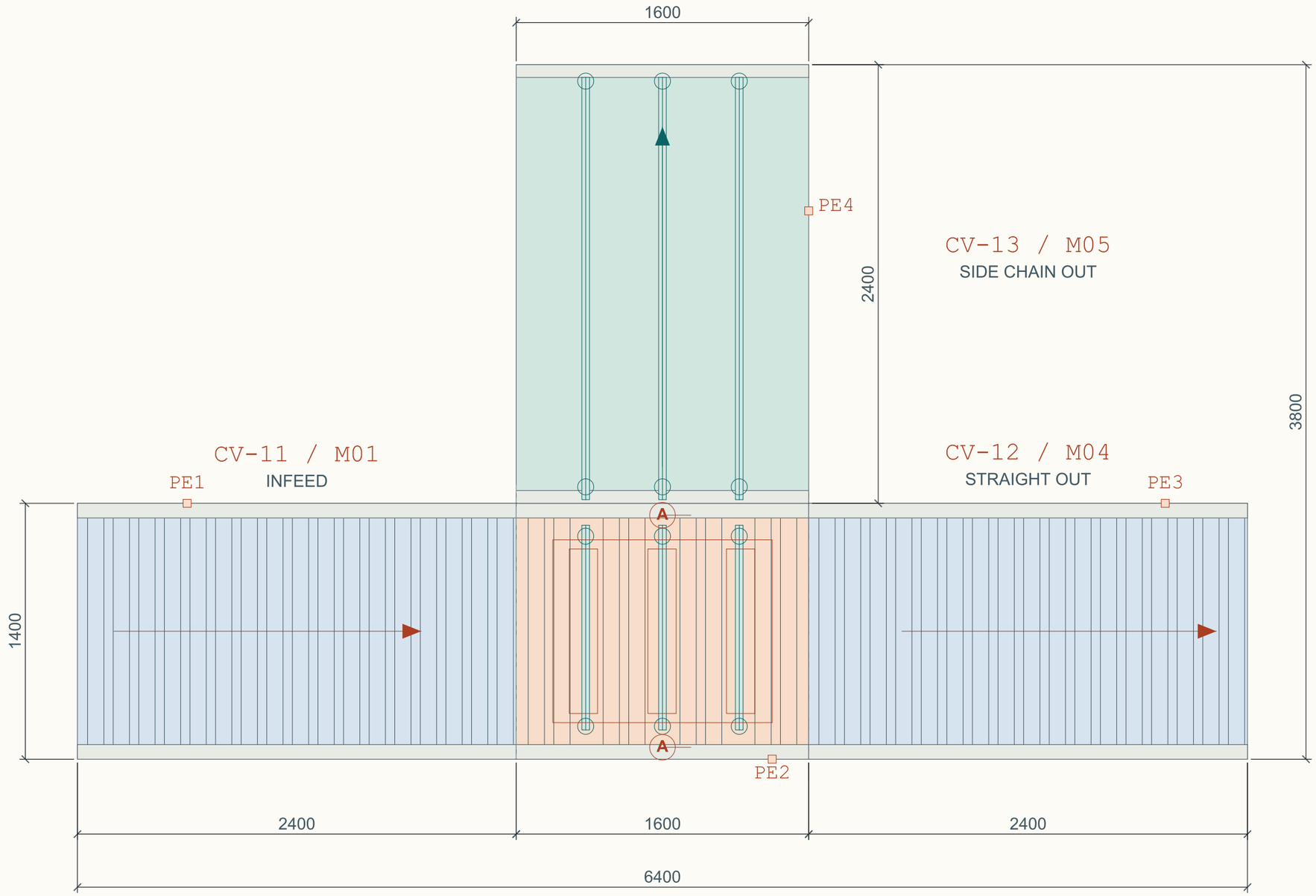
Complete drawing index on G000.

DESIGN CASE

1,200 × 1,000 mm pallet
500 kg assumed pallet mass
45 mm guided working lift

General arrangement

STRAIGHT / DIVERT ROUTES / 1,200 × 1,000 PALLET / OPEN INTERFACE DATUMS



01 TOP VIEW / MAIN + SIDE INTERFACE

MODULE REGISTER

MODULE	DRV	FUNCTION
CV-11	M01	Infeed rollers
TR-21	M02	Transfer rollers
TR-22	M03	Lift chains
CV-12	M04	Straight outfeed
CV-13	M05	Side chain outfeed

TWO TRANSFER MODES

Straight: pallet stays on rollers.
Divert: chains lift the pallet clear of the fixed roller deck.

FIELD INTERFACES

PE1 infeed / PE2 at transfer
PE3 straight out / PE4 side out
A–A section shown on M200.

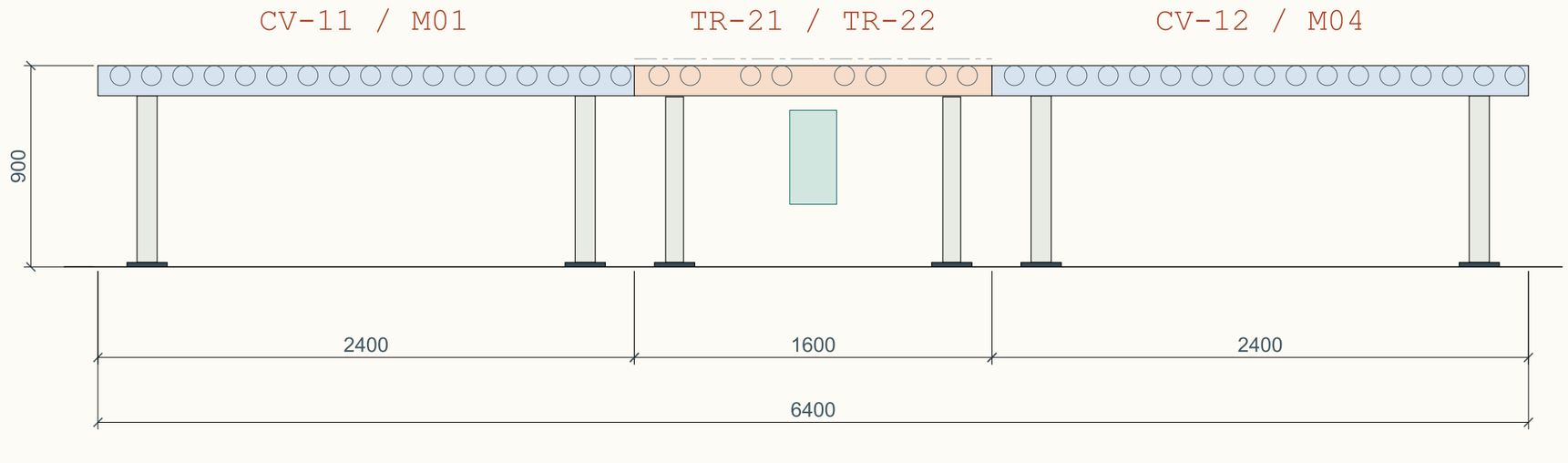
PLAN KEY

Rust rectangle: nominal pallet
Teal: chain transport route
Blue bays: live roller modules

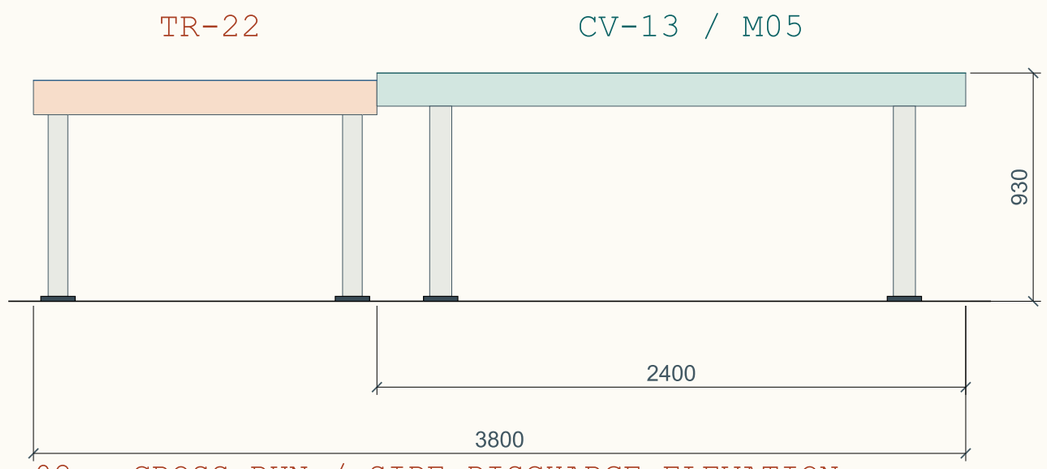
UNITS / MM

Installation elevations

DECK LEVELS / FRAME SUPPORT DATUMS / END INTERFACES



01 MAIN RUN / SIDE ELEVATION



02 CROSS RUN / SIDE DISCHARGE ELEVATION

DATUM REGISTER

SURFACE	Z (MM)
Floor / feet	0
Fixed roller top	900
Lift chains down	885
Lift chains up	930
Side chain deck	930

LEVEL TRANSFER

Raise 45 mm from chain-low.
Side deck matches chain-high.
See moving assembly on M200.

FLOOR INTERFACE

Support coordinates on M120.
Typical footplate on M220.
Confirm slab and anchor design.

INSTALLATION DATUM

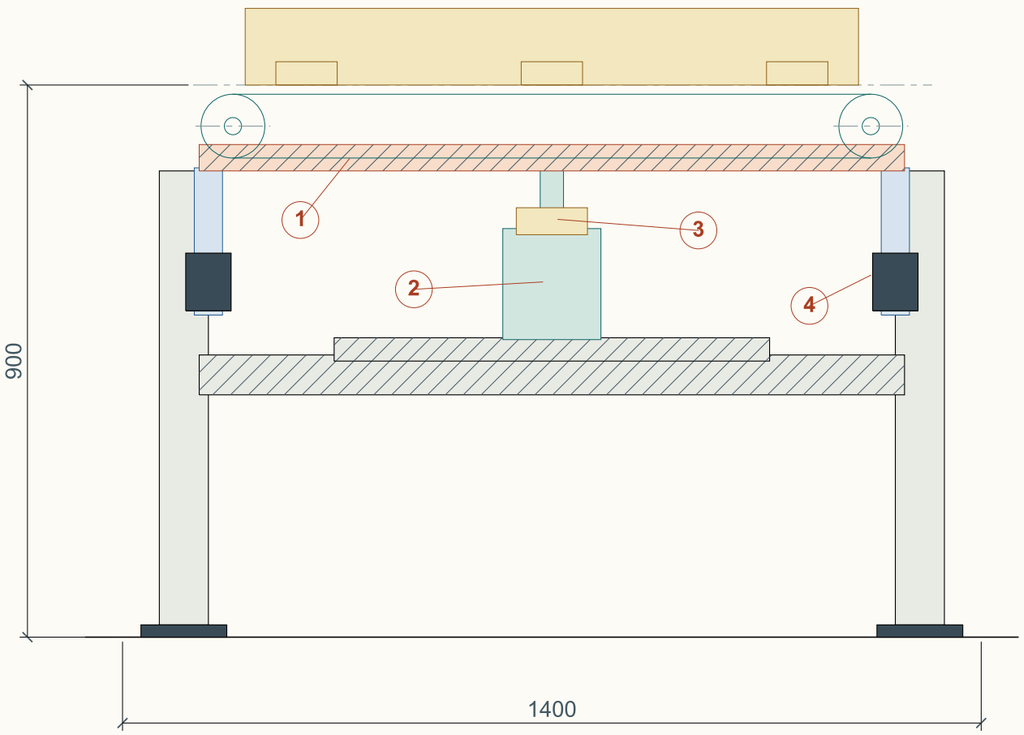
Origin at infeed / near rail.
Main transport in +X.
Side discharge in +Y.

Guided lift assembly

SECTION A–A ALONG CENTRE CHAIN STRAND / FIXED ROLLER DATUM +900

CHAIN TOP +885

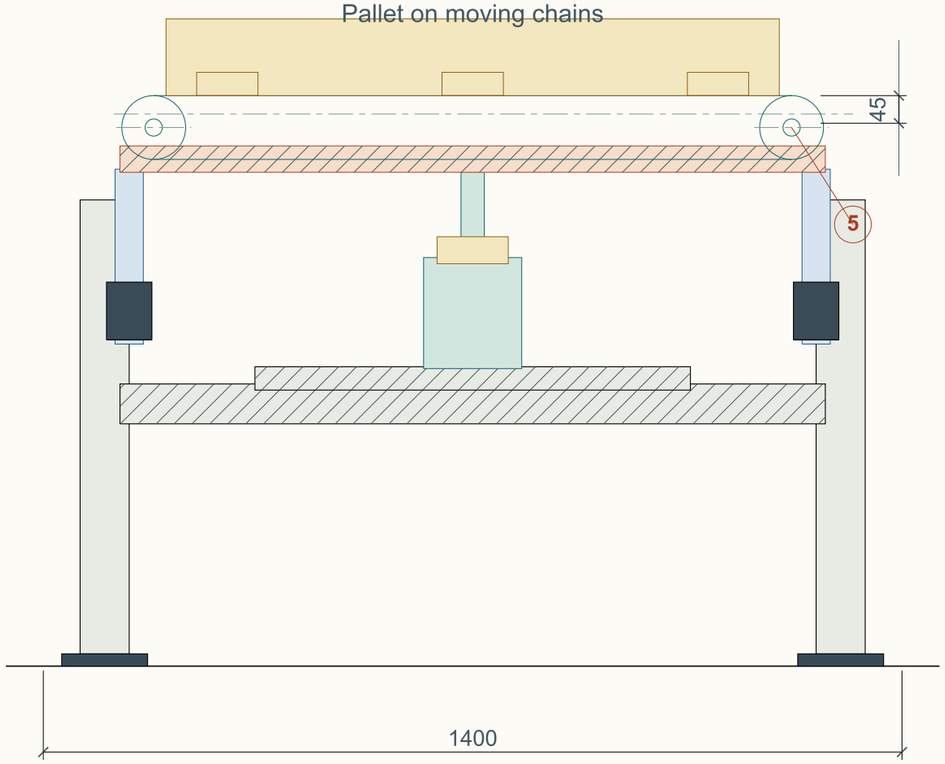
Pallet supported on fixed rollers



01 LOWERED / STRAIGHT

CHAIN TOP +930

Pallet on moving chains



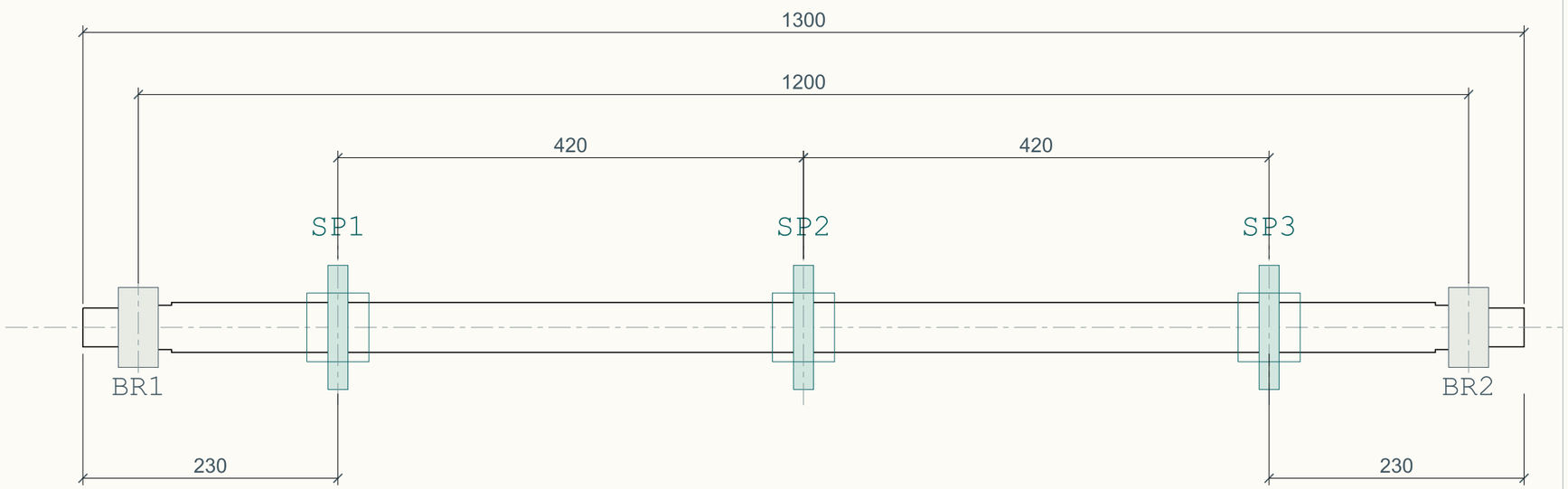
02 RAISED / TRANSFER

NO	PART	NO	PART	NO	PART	NO	PART
1	Lift carrier	2	CY1 / air cylinder	3	LK1 / rod lock	4 / 5	Guide / sprocket

A–A shows two of the four guide points projected into the section.

Transfer drive shaft

DS-01 / THREE SPROCKET STATIONS ON A COMMON SHAFT / INLINE GEARMOTOR



01 SHAFT ELEVATION / SPROCKET AND BEARING CENTRES

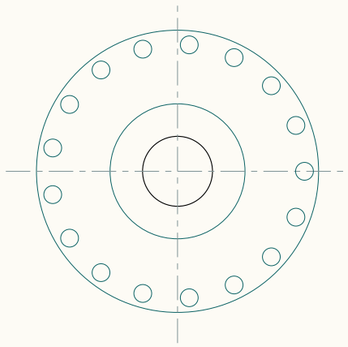
02 DRIVE INTERFACE / NTS



GEARMOTOR FLEX COUPLING DRIVE SHAFT

M03 and CPG1 travel with the lift carrier.

03 SPROCKET END / NTS



17T / PITCH CIRCLE

SHAFT PROFILE

FEATURE	DIA (MM)
Drive / tail end	35
Bearing seats	40
Sprocket body	45

CHAIN GEOMETRY

Three stations at 420 mm pitch.
17 teeth / 19.05 mm chain pitch.
Pitch diameter ≈ 103.7 mm.

KINEMATIC TARGET

Chain speed: 0.20 m/s.
Shaft target: ~37 rpm.
Size drive for starting loads.

HUB INTERFACE

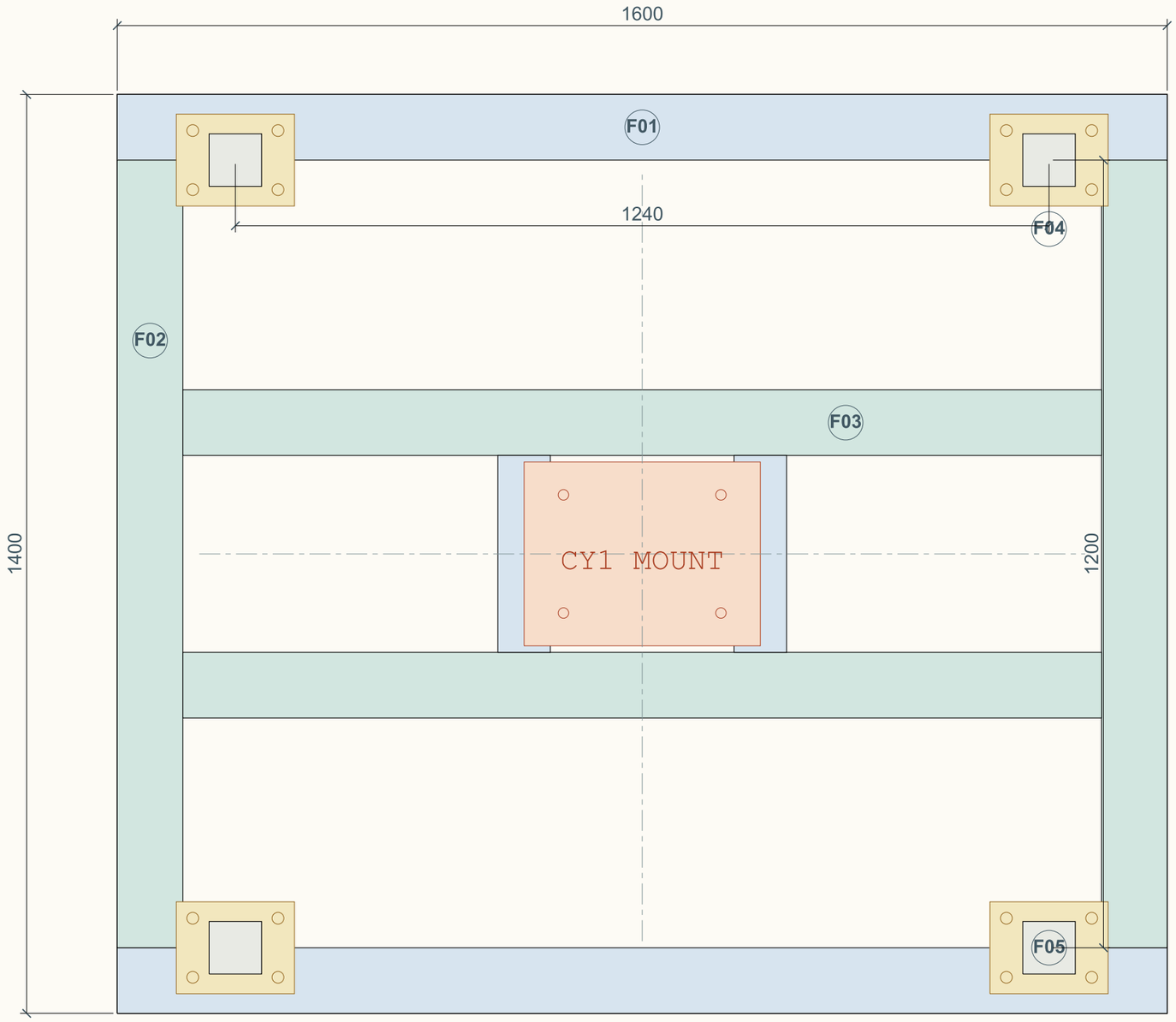
Split-clamp hubs, 3 positions.
End retention and fits to
be specified after selection.

ASSEMBLY LINKS

Lift: M200. Chain take-up: M230.

Welded base frame

WF-01 / FIXED BASE AND ANCHOR PATTERN / TYPICAL FOOTPLATE



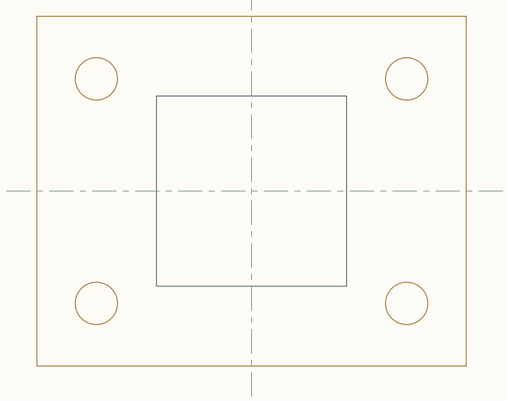
01 FRAME PLAN / TRANSFER MODULE

CUT LIST / NOMINAL

ID	SECTION / PLATE	QTY	LG (MM)
F01	RHS 100 × 60 × 4	2	1600
F02	RHS 100 × 60 × 4	2	1200
F03	RHS 100 × 60 × 4	2	1400
F04	SHS 80 × 80 × 4	4	740
F05	PL 180 × 140 × 20	4	—
F06	PL 360 × 280 × 16	1	—
F07	RHS 80 × 40 × 4	2	300

02 FOOTPLATE F05 / NTS

180 × 140 × 20

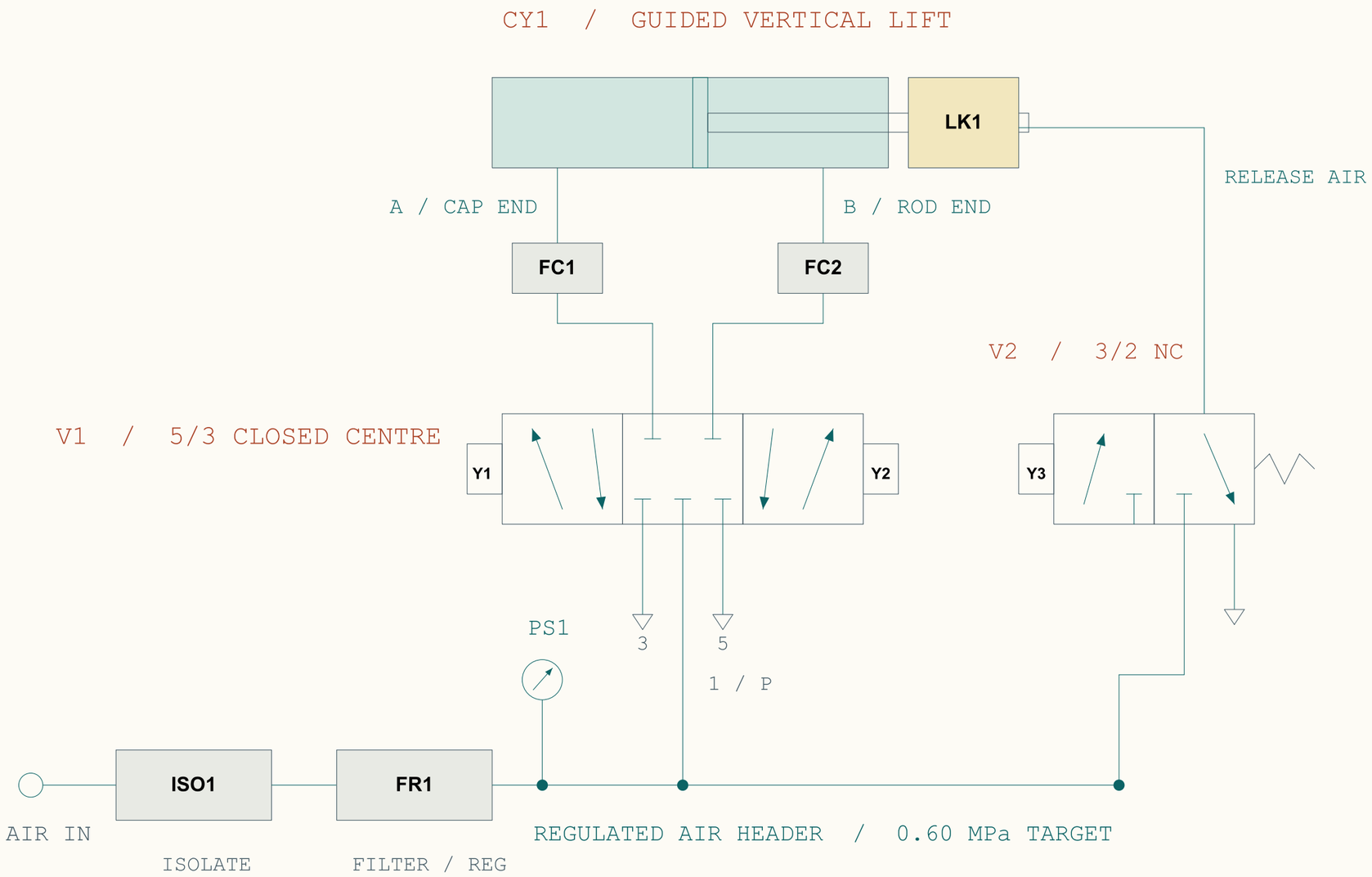


PATTERN
4 × Ø18
130 × 90
centred

ANCHOR PITCH IN PLAN
1,240 × 1,200 mm at transfer supports.

Lift pneumatics

FUNCTIONAL PNEUMATIC CIRCUIT / DOUBLE-ACTING LIFT AND SPRING-APPLIED ROD LOCK



FORCE STUDY / EXTEND

ITEM	ASSUMED
Pallet + carrier	620 kg
CY1 bore	Ø160 mm
Pressure at CY1	0.60 MPa
Static weight	6.08 kN
Ideal thrust	12.06 kN
Ideal ratio	1.98

MOTION CONTROL

FC1 / FC2: meter-out intent.
LK1 releases via V2.
PS1 proves supply pressure.

LOAD-HOLDING CONCEPT

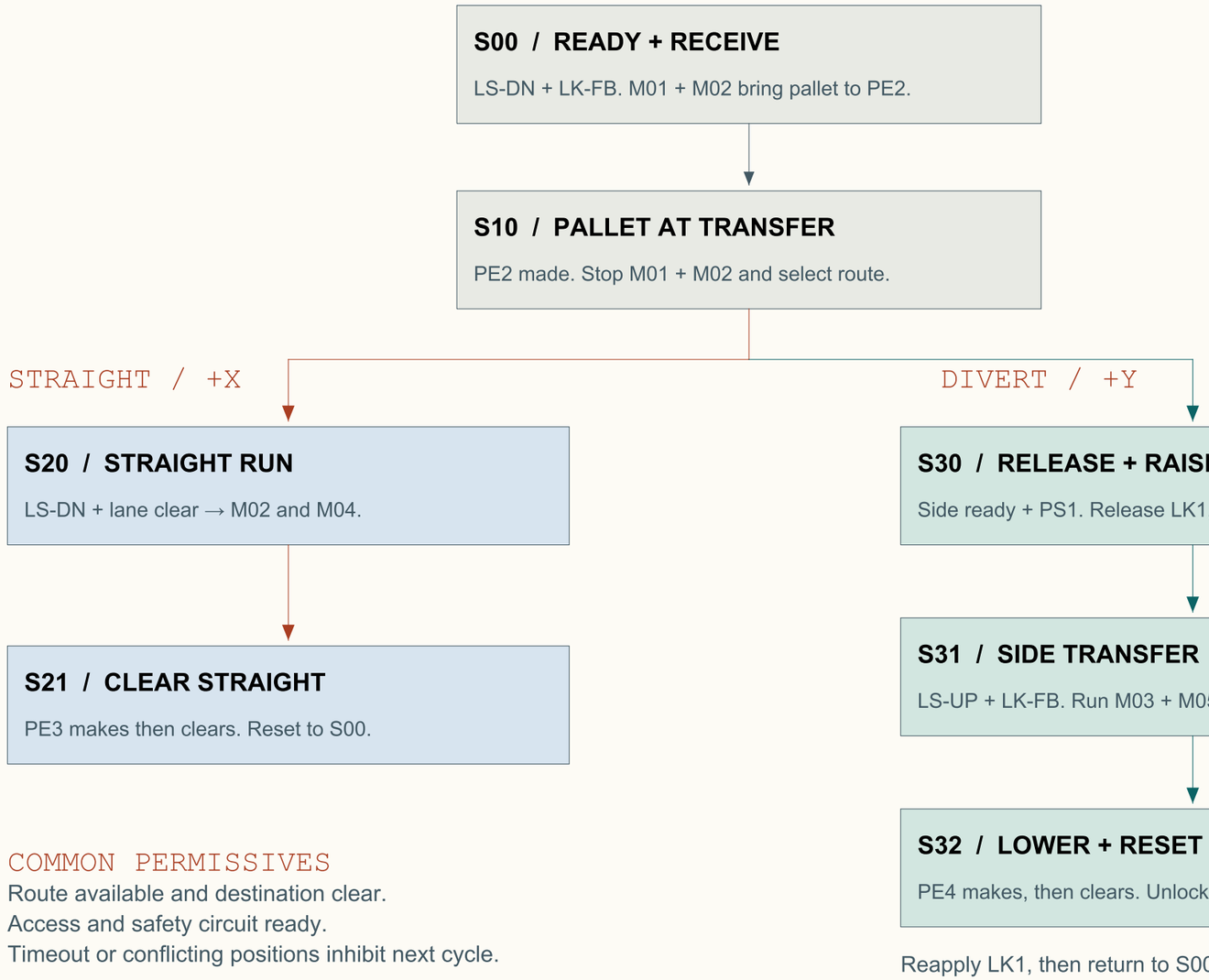
Spring-applied lock holds at the end positions. Size for load, motion and failure conditions.

DESIGN CHECKS

Verify dynamic loads, valve flow and lock capacity with the selected components.

Operating sequence

STRAIGHT AND SIDE TRANSFER STATES / POSITION CONFIRMATION AND RELEASE PERMISSIVES



Reapply LK1, then return to S00.

COMMON PERMISSIVES

Route available and destination clear.
Access and safety circuit ready.
Timeout or conflicting positions inhibit next cycle.

01 OPERATING STATES / TRANSITIONS

INPUT REGISTER

TAG	FUNCTION
PE1	Infeed present
PE2	Pallet at transfer
PE3	Straight exit
PE4	Side exit
LS-DN	Lift lowered
LS-UP	Lift raised
PS1	Pressure available
LK-FB	Locked / released feedback

OUTPUT REGISTER

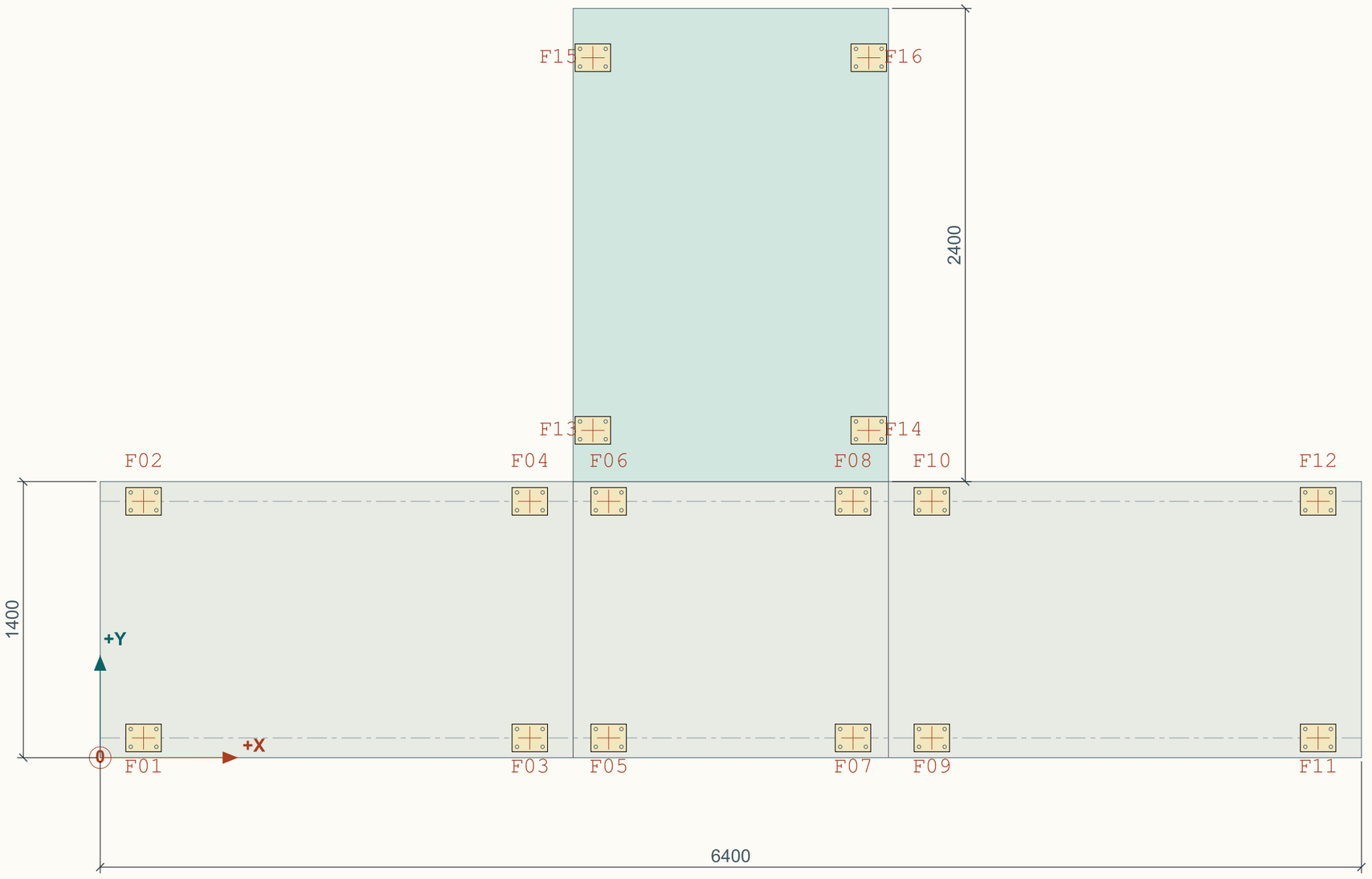
TAG	FUNCTION
M01 / M02	Infeed / transfer rollers
M03	Transfer chains
M04 / M05	Straight / side outfeed
Y1 / Y2	Lift up / down
Y3	Release LK1

CONTROL INTERFACES

Valve tags P300; controls E500.

Anchor coordinate plan

X0 AT INFEED FACE / Y0 AT NEAR RAIL / Z0 AT FINISHED FLOOR



01 ANCHOR SET-OUT / 16 SUPPORT CENTRES

CENTRE COORDINATES / MM

FOOT	X	Y
01 / 02	220	100 / 1300
03 / 04	2180	100 / 1300
05 / 06	2580	100 / 1300
07 / 08	3820	100 / 1300
09 / 10	4220	100 / 1300
11 / 12	6180	100 / 1300
13	2500	1660
14	3900	1660
15	2500	3550
16	3900	3550

TYPICAL EACH FOOT

Plate 180 × 140 × 20.
4 × Ø18 holes, centred.
Hole offsets: X ±65, Y ±45.

COORDINATION HOLD

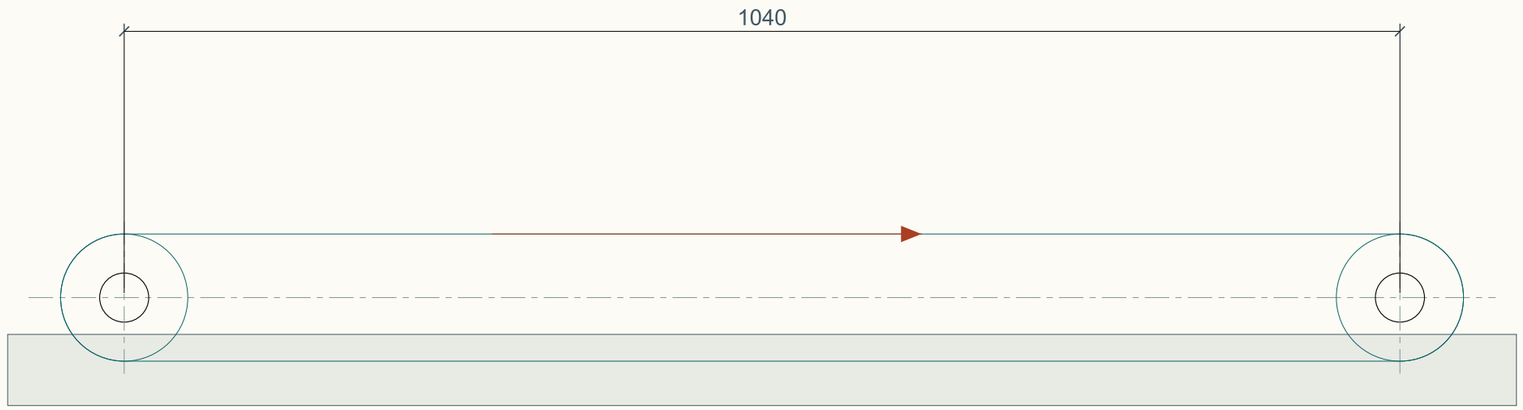
Confirm slab, anchors and operating reactions before installation release.

FOOT DETAIL / M220

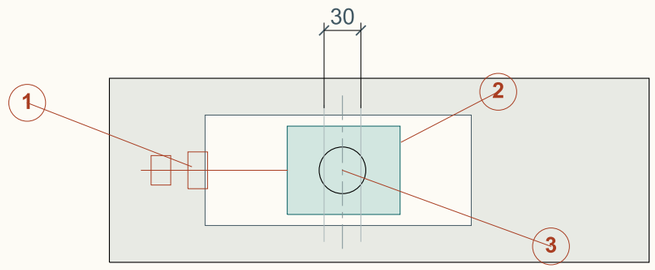
Chain and take-up assembly

TR-22 CHAIN CARRIER / PAIRED ADJUSTMENT AT THE TAIL SHAFT

01 CENTRE STRAND / NOMINAL PITCH-LINE VIEW



02 TAKE-UP BLOCK / TYPICAL BOTH SHAFT ENDS



ADJUSTMENT

Equal movement at both bearing blocks; lock after alignment.

TR-22 CHAIN GEOMETRY

ITEM	VALUE
Strands	3
Chain pitch	19.05 mm
Sprockets	17T / 17T
Centre, nominal	1,040 mm
Take-up, total	30 mm
Loop estimate	126 pitches

TAKE-UP CALLOUTS

1 Adjuster and locknut
2 Sliding bearing block
3 Common tail shaft TS-01

ALIGNMENT

Match both slide settings. Confirm chain sag and guides with the chain supplier.

DRAWING LINKS

Drive end: M210.
Carrier position: M200.

Equipment bill of materials

CV-01 / ASSEMBLY-LEVEL REGISTER / QUANTITIES PER SYSTEM

04 MODULES **05** DRIVES **42** ROLLERS **16** SUPPORTS

01 MECHANICAL ASSEMBLIES

TAG	DESCRIPTION	QTY	DRAWING
CV-11	Infeed conveyor assembly	1	M100 / M110
TR-21	Transfer roller deck	1	M100 / M200
TR-22	Guided chain carrier assembly	1	M200 / M230
CV-12	Straight outfeed assembly	1	M100 / M110
CV-13	Side chain outfeed assembly	1	M100 / M110
WF-01	Transfer base weldment	1	M220
F01–F16	Levelling support assemblies	16	M120 / M220
R-01	Live rollers, nominal Ø89	42	17 + 8 + 17
DS-01	Common chain drive shaft	1	M210
TS-01	Common tail shaft	1	M230
SP1–SP6	Lift-chain sprockets, 17T	6	3 drive + 3 tail
CH-01	Matched lift chain loops	3	M230
CH-02	Side conveyor chain loops	3	CV-13 module
TU-01	Tail slide and adjuster sets	2	M230
GP-01	Lift guide positions	4	M200

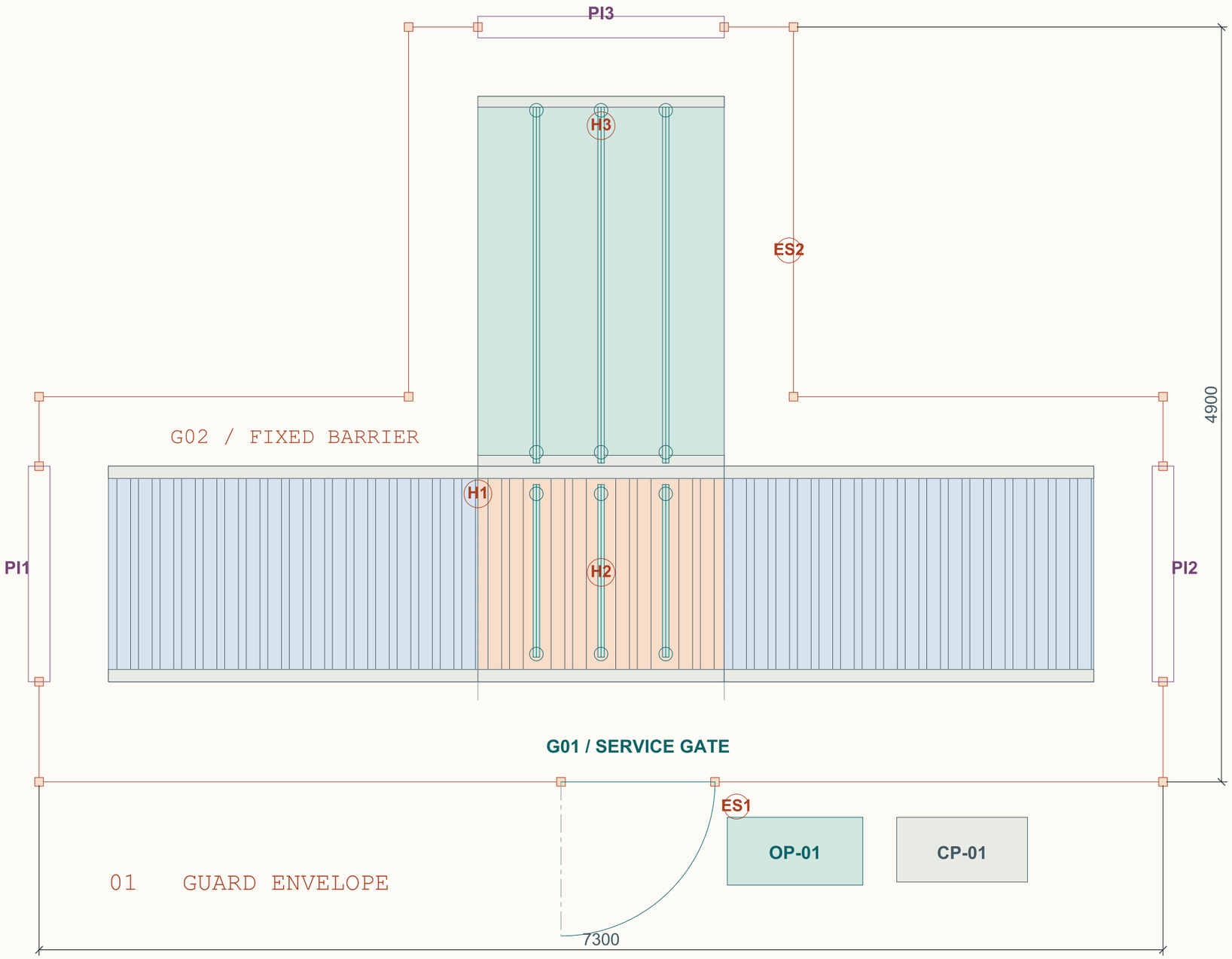
02 PURCHASE / INTEGRATION REGISTER

TAG	DESCRIPTION	QTY	DRAWING
M01–M05	Geared drive assemblies	5	M100 / E500
CPG1	Lift-chain shaft coupling	1	M210
BR1 / BR2	Drive shaft bearing units	2	M210
CY1	Double-acting lift cylinder	1	P300
LK1	Spring-applied rod lock	1	P300
V1 / V2	Lift / release valve assemblies	2	P300
FC1 / FC2	Meter-out flow controls	2	P300
ISO1 / FR1	Air isolation / preparation	2	P300
PS1	Supply pressure switch	1	P300
PE1–PE4	Pallet position sensing points	4	M100 / C400
LS-UP / DN	Lift position sensing points	2	C400 / E500
CP-01	Electrical control panel	1	E500
OP-01	Operator and reset station	1	M400 / E500
G01 / G02	Service gate / fixed guard set	2	M400
PI1–PI3	Material opening interfaces	3	M400 / E500

Assembly-level register. TR-21 and TR-22 form one physical station. Final order codes are release inputs.

Guarding and service layout

SERVICE GATE / MATERIAL OPENINGS / DRIVE AND LIFT ACCESS ZONES



HAZARD LOCATION MAP

KEY	AREA
H1	Roller / transfer nip
H2	Lift shear and crush
H3	Chain / terminal nip

G01 / SERVICE ACCESS

Gate interlock input to safety logic. Keep reset outside the enclosure with a clear view.

PI1–PI3 / LOAD PORTS

Choose the protective device and opening geometry with the upstream/downstream supplier.

SERVICE ENERGY

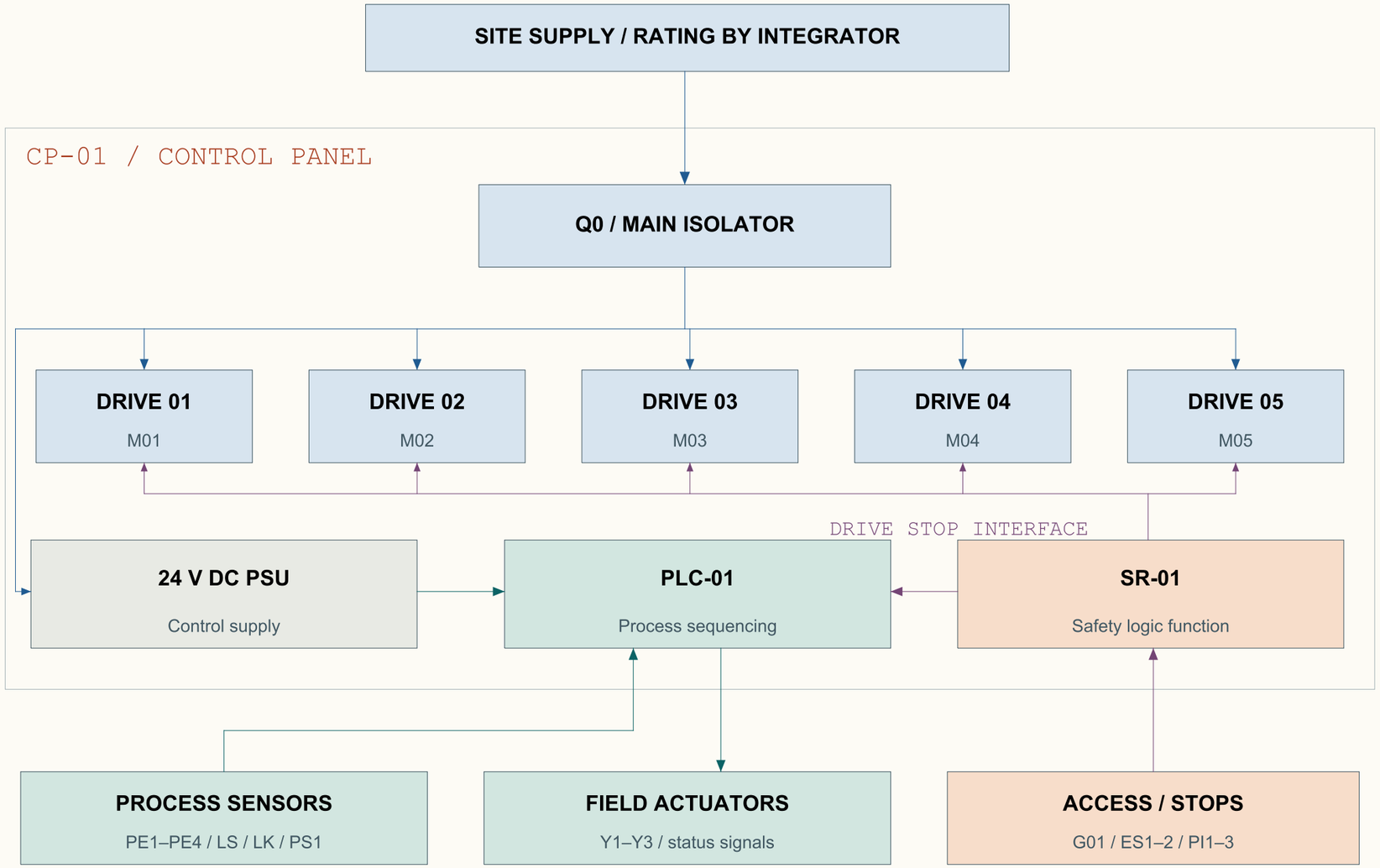
Provide lockable electrical and air isolation. Verify a rated mechanical lift restraint.

VALIDATION HOLD

Confirm reach, stopping time, access, escape and reset design. Interface diagram: E500.

Controls and field interfaces

POWER AND CONTROL BLOCK DIAGRAM / UPSTREAM AND DOWNSTREAM HANDSHAKES



CONTROL BOUNDARY

Drive stop and pneumatic load retention are separate safety design inputs.

LINE HANDSHAKES

SIGNAL	DIRECTION
IN_READY	Cell → upstream
IN_REQ	Upstream → cell
ROUTE_SEL	Line → cell
OUT_X_RDY	Straight → cell
OUT_Y_RDY	Side → cell
OUT_DONE	Cell → line
FAULT	Cell → line

TRANSFER PERMISSION

Require selected outfeed ready and route clear before release. Timeout holds the next cycle.

RESTART POLICY

Reset acknowledges the trip. Issue a separate start command after all permissives recover.

RELEASE INPUTS

Confirm supply, drive rating, wiring and safety performance. Validate stop / hold behaviour.

Verification and handover

FACTORY AND SITE ACCEPTANCE PLAN / MEASUREMENT RECORDS / RELEASE HOLD POINTS

01 PROPOSED ACCEPTANCE MATRIX

ID	STAGE	CHECK	RECORD / ACCEPTANCE INTENT	REF
01	FAT	Identification	Verify module, motor and field device tags against the issue register.	M100 / M300
02	SAT	Installation set-out	Measure the 16 foot centres and record actual floor and deck elevations.	M110 / M120
03	FAT	Chain tracking	Record both take-up positions and verify tracking over a complete unloaded loop.	M210 / M230
04	FAT	Lift positioning	Measure +885 and +930 chain levels. Verify the 45 mm working stroke and feedback.	M200 / P300
05	FAT	Straight cycle	Record PE1 → PE2 → PE3 state order and straight outfeed handshake.	C400 / E500
06	FAT	Divert cycle	Record lift, lock feedback, side discharge and return to the lowered datum.	C400 / E500
07	FAT / SAT	Access and restart	Under the approved test plan, check protective trips, reset and separate start.	M400 / E500
08	FAT / SAT	Energy loss / restore	Verify stop, lift load retention and restart behaviour against approved criteria.	P300 / E500
09	SAT	Loaded cycle	Run the agreed test pallet and load after guarding, anchorage and controls are accepted.	G000 / C400

02 ENGINEERING RELEASE HOLD POINTS

A / MECHANICAL
Confirm load spectrum, fatigue, chain pull, shaft fits, welds and floor reactions.

B / COMPONENTS
Select motors, bearings, cylinders, lock and valves from confirmed duty conditions.

C / SAFETY + SITE
Assess access and stopping risks. Validate guards, interlocks and energy isolation.

FAT = factory acceptance test / SAT = site acceptance test / Record actual values in the project test report.